Arista Access Router JAY-ACC-47 Routing Protocol Flap Recovery SOP

Node: JAY-ACC-47 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_6183

Description

In the context of routing protocol flap failures within the Arista access router JAY-ACC-47, deployed at the Jayanagar Metro Site, a major alarm (ALARM_6183) is triggered due to the loss of BGP routes, OSPF adjacency, or ISIS LSP flooding. This phenomenon occurs due to the inherent complexity of routing protocols interacting with the underlying hardware architecture of the Arista 7280R access router. Specifically, the routing protocol flap phenomenon is characterized by the rapid oscillation of routing table updates between routers, resulting from the dynamic exchange of routing information. This oscillation can be exacerbated by factors such as high link utilization, network congestion, or misconfigured routing policies. Furthermore, the Arista access router's hardware architecture, comprising of multiple routing engines and switching fabrics, can contribute to the routing protocol flap phenomenon by introducing latency and jitter in the processing of routing updates. Consequently, the routing protocol flap failure in the context of the Arista access router JAY-ACC-47 necessitates immediate attention to prevent network instability and minimize downtime.

Pre-requisite Checks

- Verify that the JAY-ACC-47 access router is powered on and operational.
- Confirm that the redundant link is online and functioning correctly.
- Dispatch remote hands to the Jayanagar Metro Site to ensure physical access to the equipment.

Potential Root Causes

Cause	Probability
BGP route churn caused by high link utilization	High
OSPF adjacency loss due to misconfigured routing policies	Medium
ISIS LSP flooding caused by network congestion	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version 1, local router ID 192.168.1.1, status 3, table version 1, BGP routing table entries 10, peered with 2 neighbor, Description: This command displays the BGP routing table and helps identify any churn or oscillation in the routing table updates.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays the OSPF neighbor relationships and helps identify any adjacency loss or instability.
```

```
Command: show isis database, Expected Output: IS-IS level 1 database, LSP ID 1.1.1.1.1, sequence number 0x12345678, Description: This command displays the ISIS database and helps identify any LSP flooding or instability.
```

Resolution Steps

Step 1: Step: Clear BGP routing table and restart BGP process, Command: clear bgp *, Description: This command clears the BGP routing table and restarts the BGP process to resolve the routing protocol flap phenomenon.

Step 2: Step: Verify OSPF adjacency and restart OSPF process if necessary, Command: show ospf neighbor; restart ospf, Description: This command verifies the OSPF adjacency and restarts the OSPF process if necessary to resolve the adjacency loss.

Step 3: Step: Verify ISIS LSP flooding and restart ISIS process if necessary, Command: show isis database; restart isis, Description: This command verifies the ISIS LSP flooding and restarts the ISIS process if necessary to resolve the LSP flooding.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version 1, local router ID 192.168.1.1, status 3, table version 1, BGP routing table entries 10, peered with 2 neighbor, Description: This command verifies that the BGP routing table is stable and free of churn.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command verifies that the OSPF adjacency is stable and active.

Command: show isis database, Expected Output: IS-IS level 1 database, LSP ID 1.1.1.1.1, sequence number 0x12345678, Description: This command verifies that the ISIS LSP flooding is resolved and the database is stable.
```

Roll-back Procedures

- Step: Restore the original routing configuration, Command: copy running-config startup-config, Description: This command restores the original routing configuration and resolves the routing protocol flap phenomenon.

- Step: Verify the routing protocol stability, Command: show bgp summary; show ospf neighbor; show isis database, Description: This command verifies that the routing protocol stability is restored and the network is operational.